

LAKSHYA

JEE 2027

Lecture No.- 15

MATHEMATICS

**LIMIT, CONTINUITY &
DIFFERENTIABILITY**

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Topics *to be covered*



1 Algebra or Properties of Differentiability

2 Important Results

3 Question Practice

4





LAST CLASS RECAP

Answer the following Questions:

Range of cont. funcⁿ over closed interval
is also closed interval.

(i) If $f(x)$ is a continuous function & $f: [3, 4] \rightarrow (0, 5)$, then $f(x)$ can be onto function.

(ii) If $f(x)$ is a continuous function & $f: [0, 1] \rightarrow \mathbb{Q}$ with $f\left(\frac{3}{5}\right) = \frac{5}{3}$,
then $f(0) + f\left(\frac{1}{2}\right) + f\left(\frac{1}{4}\right) = 5$. $\textcircled{T}$

Handwritten notes for (ii):
- $f(x) = \frac{5}{3}$ (with a pink arrow pointing to the value 5/3)
- Four yellow stars next to the value 5/3.
- A green arrow points from the value 5/3 to a green circle containing 'F'.

(iii) If a function is discontinuous at $x = a$, then it is non-differentiable at $x = a$. $\textcircled{T}$

(iv) Calculating $f'(x)$ to check differentiability at a point is always applicable.

Handwritten note for (iv):
- A yellow arrow points from the end of the sentence to a yellow circle containing 'F'.

$f: [0,1] \rightarrow \mathbb{Q}$
 ↪ continuous

Constant funcⁿ.





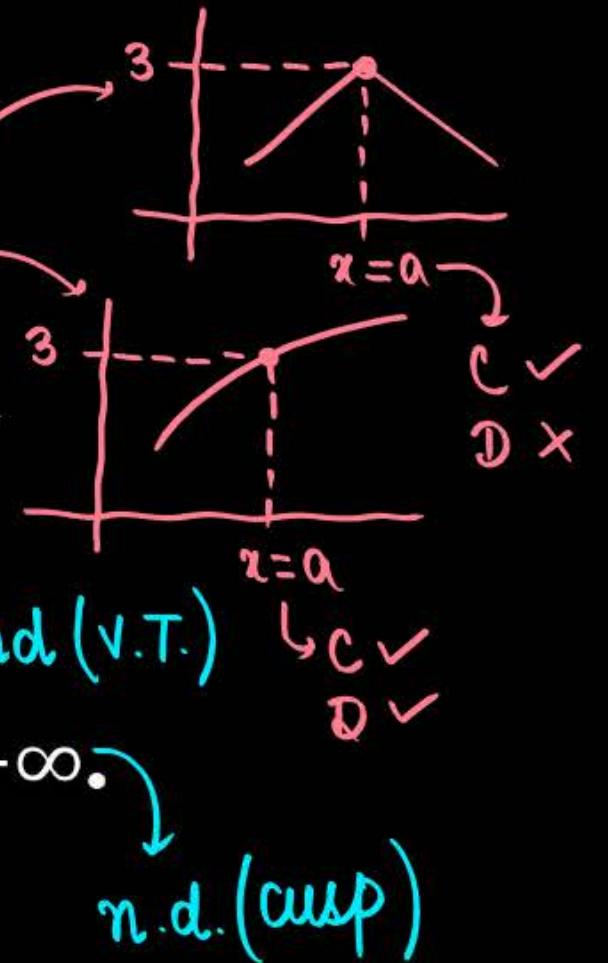
LAST CLASS RECAP

$\mathcal{D}$

$\mathcal{N.D.}$

In the following Situations, $f(x)$ is **differentiable** or **non-differentiable** at $x = a$??

- (i) at $x = a$: $\text{LHL} = -4$ & $\text{RHL} = 3 \rightarrow \mathcal{N.D.}$ (because $\mathcal{D.C.}$)
- (ii) at $x = a$: $\text{LHL} = \text{RHL} = 3$ & $f(a) = 2 \rightarrow \mathcal{N.D.}$
- (iii) at $x = a$: $\text{LHL} = \text{RHL} = f(a) = 3 \rightarrow \text{can't comment } (\mathcal{D.} \text{ OR } \mathcal{N.D.})$
- (iv) at $x = a$: $\text{RHD} = 2$ & $\text{LHD} = -2 \rightarrow \mathcal{N.D.}$ (Sharp corner) but continuous
- (v) at $x = a$: $\text{RHD} = 3 = \text{LHD.} \rightarrow \mathcal{D.}$
- (vi) at $x = a$: function is continuous and $\text{LHD}, \text{RHD} \rightarrow +\infty. \rightarrow \mathcal{N.D.} (\text{V.T.})$
- (vii) at $x = a$: function is continuous and $\text{LHD} \rightarrow +\infty, \text{RHD} \rightarrow -\infty.$



$\mathcal{N.D.}$ (cusp)

H.W. Discussion

The left hand derivative of $f(x) = [x] \sin(\pi x)$ at $x = k$, k is an integer, is

(Given: $[.] = \text{Gif}$)

A $(-1)^k (k-1)\pi$

B $(-1)^{k-1} (k-1)\pi$

C $(-1)^k k\pi$

D $(-1)^{k-1} k\pi$

$$dH.D. = \lim_{h \rightarrow 0^+} \frac{f(k-h) - f(k)}{-h}$$

$$\lim_{h \rightarrow 0^+} \frac{[k-h] \sin(\pi(k-h))}{-h}$$

$$\frac{(k-1)(-1)^{k-1} \sin \pi h}{-\pi h} \times \pi$$

$$dH.D. = (k-1)(-1)^k \pi$$

$$\sin(\pi k - \pi h)$$

$k = \text{odd}$

$k = \text{even}$

$$+\sin \pi h$$

$$(-1)^{k+1} \text{ or } (-1)^{k-1}$$

$$-\sin \pi h$$

$$(-1)^{k+1} \text{ or } (-1)^{k-1}$$

Let k and m be positive real numbers such that the function

$$f(x) = \begin{cases} 3x^2 + k\sqrt{x+1}, & 0 < x < 1 \\ mx^2 + k^2, & x \geq 1 \end{cases} \text{ differentiable for all } x > 0. \text{ Then } \frac{8f'(8)}{f'(\frac{1}{8})} \text{ is}$$

equal to

cont. at $x=1$

$$\text{LHL} = 3 + \sqrt{2}k$$

$$\text{RHL} = f(1) = m + k^2$$

$\Downarrow$

$$3 + \sqrt{2}k = m + k^2$$

$$f'(x) = 6x + k \times \frac{1}{2} (x+1)^{-\frac{1}{2}}, \quad 0 < x < 1$$

$$= 2mx, \quad x > 1$$

$$f'(1^+) = f'(1^-)$$

$$2m = 6 + \frac{k}{2\sqrt{2}}$$

$$f'(8) = 16m$$

$$f'(\frac{1}{8}) =$$

$$m + k^2 = 3 + \sqrt{2}k$$

$$\cancel{3} + \frac{k}{4\sqrt{2}} + k^2 = \cancel{3} + \sqrt{2}k$$

$$6 + \frac{k}{2\sqrt{2}} = 2m$$

$$k \left(\frac{1}{4\sqrt{2}} + k - \sqrt{2} \right) = 0$$

$$\begin{aligned} & k = 0 \\ & \Downarrow \\ & m = 3 \end{aligned}$$

Rejected.

$$k = \sqrt{2} - \frac{1}{4\sqrt{2}}$$

$$k = \frac{7}{4\sqrt{2}}$$

$$m = 3 + \frac{1}{4\sqrt{2}} \times \frac{7}{4\sqrt{2}} = 3 + \frac{7}{32} = \frac{103}{32} = m$$

$$f'(8) = 2m \times 8$$

$$= 16m$$

$$k = \frac{7}{4\sqrt{2}}$$

$$m = \frac{103}{32}$$

$$f'\left(\frac{1}{8}\right) = 6 \times \frac{1}{8} + \frac{k(2\sqrt{2})}{2(3)}$$

$$\frac{8f'(8)}{f'\left(\frac{1}{8}\right)} =$$

$$f \eta f = \frac{\overset{4}{\cancel{8}} \times \cancel{16} \times 103}{\cancel{32} \left(\frac{\cancel{6}^3}{\cancel{8}_4} + \frac{1}{6} \times \frac{7}{\cancel{4\sqrt{2}}_2} \times \cancel{2\sqrt{2}} \right)} = \frac{\cancel{4} \times 103}{\left(\frac{\cancel{4}}{3} \right)} = \boxed{309}$$



If $f(x) = \begin{cases} |x|^m \sin(\frac{1}{x}) & x \neq 0 \\ 0, & x = 0 \end{cases}$, then find the set of value of m for which

(i) $f(x)$ is continuous but not differentiable

(ii) $f(x)$ is continuous and differentiable # at $x=0$.

(iii) $f(x)$ is discontinuous

Cont. at $x=0$.

$$\text{dHL} = |0^-|^m \sin \frac{1}{0^-}$$

$$\text{RHL} = |0^+|^m \sin \frac{1}{0^+}$$

$$f(0) = 0$$

for continuous

$$m > 0$$

for diff: $m-1 > 0$

$$m > 1$$

$$\text{RHD} = \lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h}$$

$$\lim_{h \rightarrow 0^+} \frac{(h)^m \sin \frac{1}{h}}{h} = \lim_{h \rightarrow 0^+} (h)^{m-1} \sin \frac{1}{h}$$

$$\text{dHD} = \lim_{h \rightarrow 0^+} \frac{f(-h) - f(0)}{-h} = \lim_{h \rightarrow 0^+} \frac{(-h)^m \sin(\frac{1}{-h})}{-h}$$

$$\lim_{h \rightarrow 0^+} \frac{(-h)^m \sin(\frac{1}{-h})}{-h} = \lim_{h \rightarrow 0^+} (-h)^{m-1} \sin(\frac{1}{-h})$$

$$h^{m-1} \sin \frac{1}{h}$$

QUESTION



Let a and c be real numbers such that $c > 0$ and $f(x)$ is defined on $[0, 1]$ as

$$f(x) = \begin{cases} x^a \sin(x^{-c}), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$$

Show that

- (i) $f(x)$ is continuous if and only if $a > 0$
- (ii) $f'(0)$ exists if and only if $a > 1$
- (iii) $f'(x)$ is continuous if and only if $a > 1 + c$
- (iv) $f''(0)$ exists if and only if $a > 2 + c$





If $f(x)$ is differentiable at $x = a$ then $f'(x)$ need not to be always continuous at $x = a$.



example:



$$\# \left. \begin{aligned} f(x) &= x^2 \sin \frac{1}{x}, \quad x \neq 0 \\ &= 0, \quad x = 0. \end{aligned} \right\}$$

cont. & diff. at $x=0$ ✓

but $f'(0^+) \neq f'(0^-)$

oscillatory type (d.n.e.)

$$f'(x) = 2x \sin \frac{1}{x} + x^2 \cos \frac{1}{x} \cdot \left(-\frac{1}{x^2}\right), \quad x \neq 0.$$

$$RHD = f'(0^+) = \lim_{x \rightarrow 0^+} \left(\underbrace{2x}_{\downarrow 2 \cdot 0^+ \times [-1,1] \uparrow} \underbrace{\sin \frac{1}{x}}_{\downarrow [-1,1] \uparrow} - \cos \frac{1}{x} \right) = \text{D.N.E. (oscillatory limit)}$$

$$\underbrace{2 \cdot 0^+ \times [-1,1] \uparrow}_{\downarrow 0^+} - [-1,1] \uparrow$$

Similarly LHD = D.N.E. (oscillatory limit)



RELATIONSHIP BETWEEN C & D



Continuous



Differentiable

Differentiable



Continuous

Discontinuous



Non-differentiable

Non-differentiable



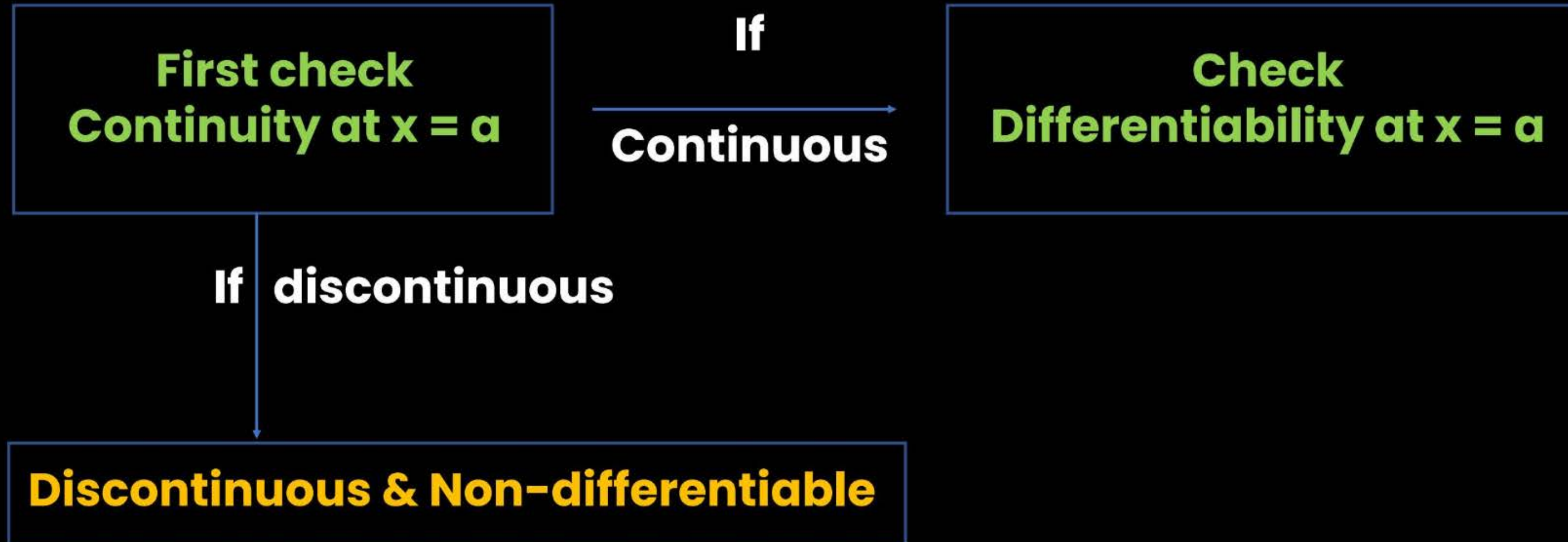
Discontinuous

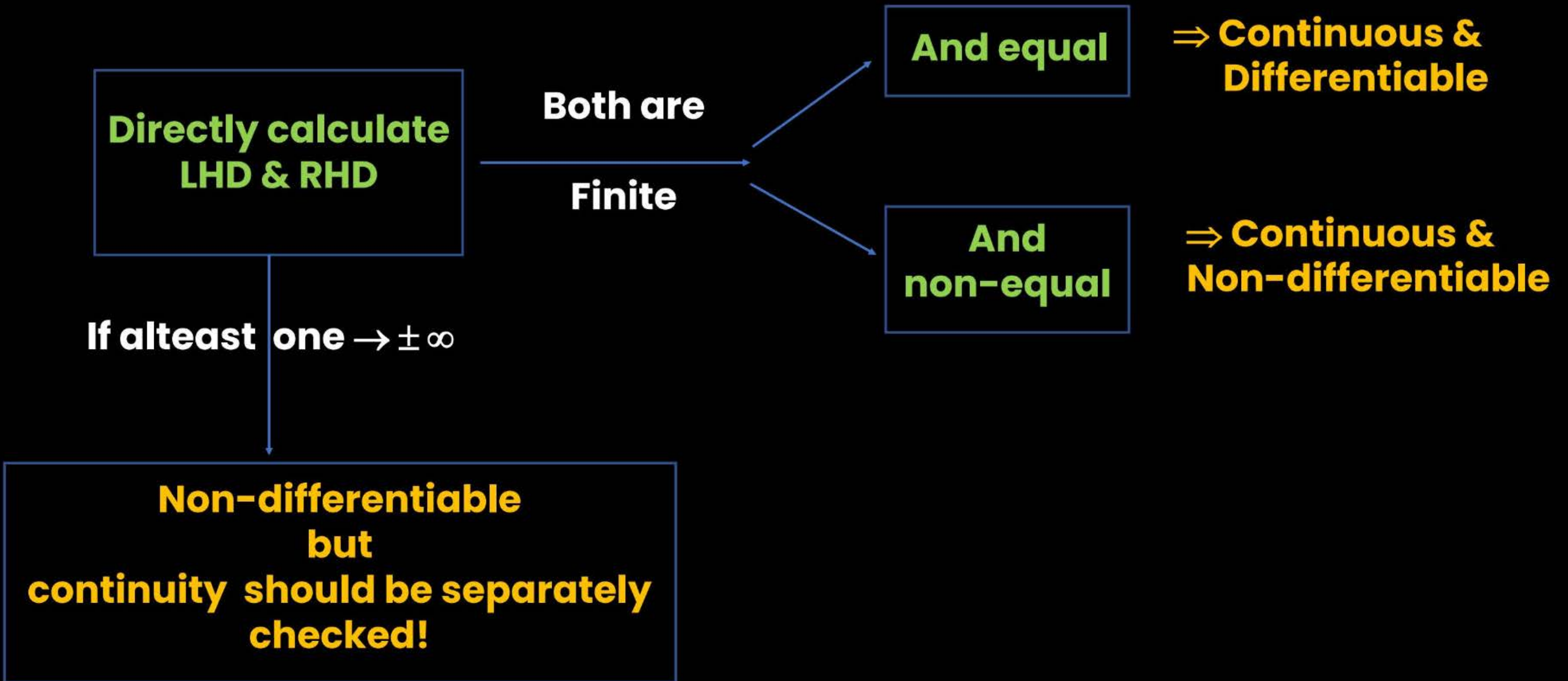
L.H.D. & R.H.D. both are finite $\Rightarrow$ Continuous



at $x=a$:

$$\begin{aligned}
 \text{RHD} &= \lim_{h \rightarrow 0^+} \frac{\overbrace{f(a+h)}^{\text{RHL}} - f(a)}{\underbrace{h}_{\text{OH}}} = \text{finite} \Rightarrow \underbrace{f(a+h)}_{\substack{\downarrow \\ \text{RHL}}} = \underbrace{f(a)}_{\downarrow} \\
 & \hspace{15em} \text{RHL} = f(a)
 \end{aligned}$$







$$f(x) = \begin{cases} (\ln(e|[x]| + [-x]))^x \cdot \left(\frac{2e^{\left(\frac{[x]+[-x]}{|x|}\right)} - 5}{3 + e^{\frac{1}{|x|}}} \right), & \text{for } x < 0 \\ 0 & \text{for } x = 0 \\ x \cdot \left(\frac{1 - e^{|x|+\{x\}}}{|x| + \{x\}} \right), & \text{for } x > 0 \end{cases}$$

Where $[.]$, $\{.\}$ represents integral and fractional part functions, respectively. Compute the Right hand derivative and Left hand derivative at $x = 0$ and comment on the continuity and derivability at $x = 0$

$$f(x) = \begin{cases} (\ln(e|[x]| + [-x])^x) \cdot \left(\frac{2e^{\left(\frac{[x]+[-x]}{|x|}\right)} - 5}{3 + e^{\frac{1}{|x|}}} \right), & \text{for } x < 0 \\ 0 & \text{for } x = 0 \\ x \cdot \left(\frac{1 - e^{|x|+\{x\}}}{|x| + \{x\}} \right), & \text{for } x > 0 \end{cases}$$

at $x=0$

$$\text{LHD} = \lim_{h \rightarrow 0^+} \frac{f(-h) - \cancel{f(0)}}{-h} = \lim_{h \rightarrow 0^+} \frac{\cancel{f(0)} \rightarrow 0}{-h}$$

$$\frac{\ln(e^{\overbrace{[-h]}^{-1}} + \overbrace{[h]}^0)^{-h}}{-h} \times \left(\frac{2e^{\frac{\overbrace{[-h]}^{-1} + \overbrace{[h]}^0}{|-h|}} - 5}{3 + e^{\frac{1}{|-h|}}} \right)$$

$$\frac{\cancel{(-h)} \ln(e)}{\cancel{-h}} \times \left(\frac{2e^{\left(\frac{-1}{h}\right)} - 5}{3 + e^{\frac{1}{h}}} \right) = 1 \times \left(\frac{2 \times 0 - 5}{3 + e^{\infty}} \right) = 0$$

$$f(x) = \begin{cases} (\ln(e|[x]| + [-x])^x) \cdot \left(\frac{2e^{\left(\frac{[x]+[-x]}{|x|}\right)} - 5}{3 + e^{\frac{1}{|x|}}} \right), & \text{for } x < 0 \\ 0 & \text{for } x = 0 \\ x \cdot \left(\frac{1 - e^{|x|+\{x\}}}{|x| + \{x\}} \right), & \text{for } x > 0 \end{cases}$$

RHD = -1, LHD = 0

Cont. at $x=0$. ✓

non-diff. at $x=0$ ✓

at $x=0$

$$\begin{aligned} \text{RHD} &= \lim_{h \rightarrow 0^+} \frac{f(h) - \cancel{f(0)}}{h} = \lim_{h \rightarrow 0^+} \frac{\cancel{h} \left(\frac{1 - e^{|h|+\{h\}}}{|h| + \{h\}} \right)}{\cancel{h}} \\ &= \lim_{h \rightarrow 0^+} \left(\frac{1 - e^{h+h}}{h+h} \right) = \frac{1 - e^{2h}}{2h} \\ &= \boxed{-1} \end{aligned}$$



DIFFERENTIABILITY OVER INTERVAL

1. If $x \in (a, b) \Rightarrow$ function should be differentiable $\forall (a < x < b)$
2. If $x \in [a, b] \Rightarrow$ function should be differentiable $\forall (a < x < b)$
 - $\Rightarrow$ At $x = a \Rightarrow$ **RHD = finite.** ✓
 - $\Rightarrow$ At $x = b \Rightarrow$ **LHD = finite.** ✓

NOTE: SUSPICIOUS POINTS IS AS SAME AS CONTINUITY ✓

Find number of points for which given functions are non-derivable :

(i) $h(x) = \min.\{x, x^2\}$ [IIT-JEE-1998]

(ii) $f(x) = \max.\{x, x^3\}$ [IIT-JEE-2001]

(iii) $g(x) = \max.\{ |\tan x|, \cos|x| \}, x \in (-\pi, \pi)$

**(iv) $h(x) = \max.\{|x| + 1, x^2 + 1\}, x \leq 0$
 $= \min.\{|x| + 1, x^2 + 1\}, x > 0$ [JEE (Adv.)-2014]**

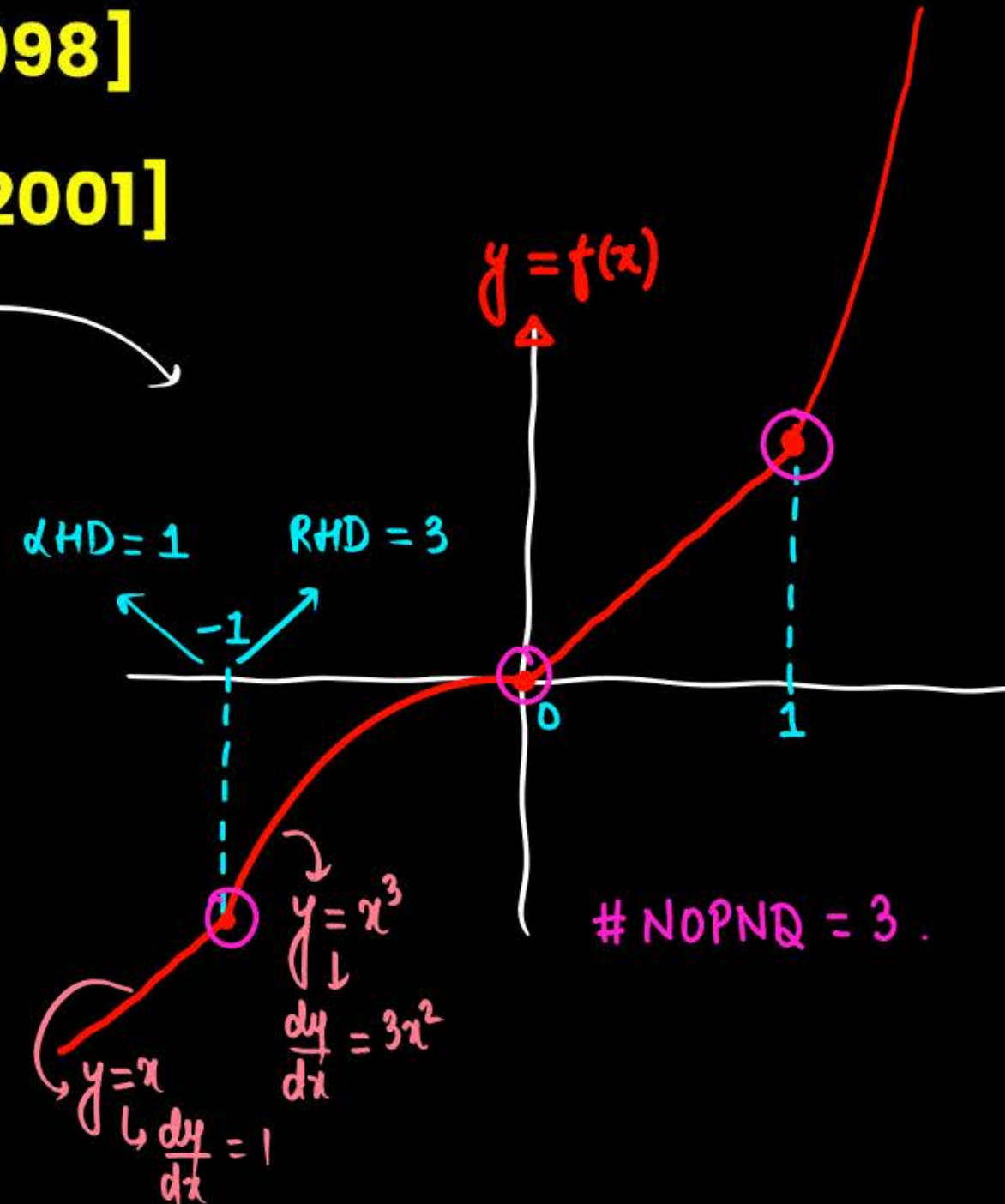
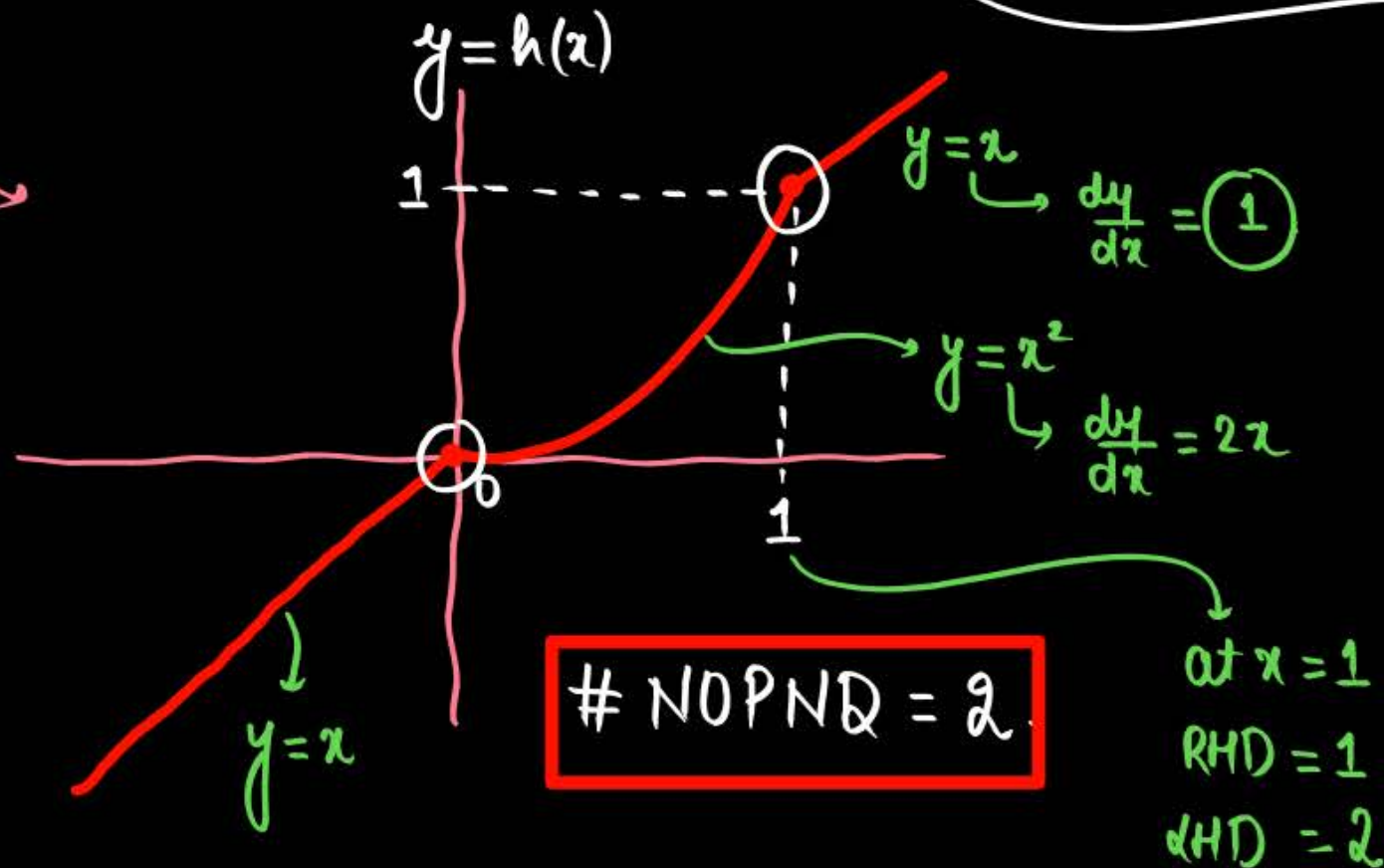
Find number of points for which given functions are non-derivative:

(i) $h(x) = \min.\{x, x^2\}$

[IIT-JEE-1998]

(ii) $f(x) = \max.\{x, x^3\}$

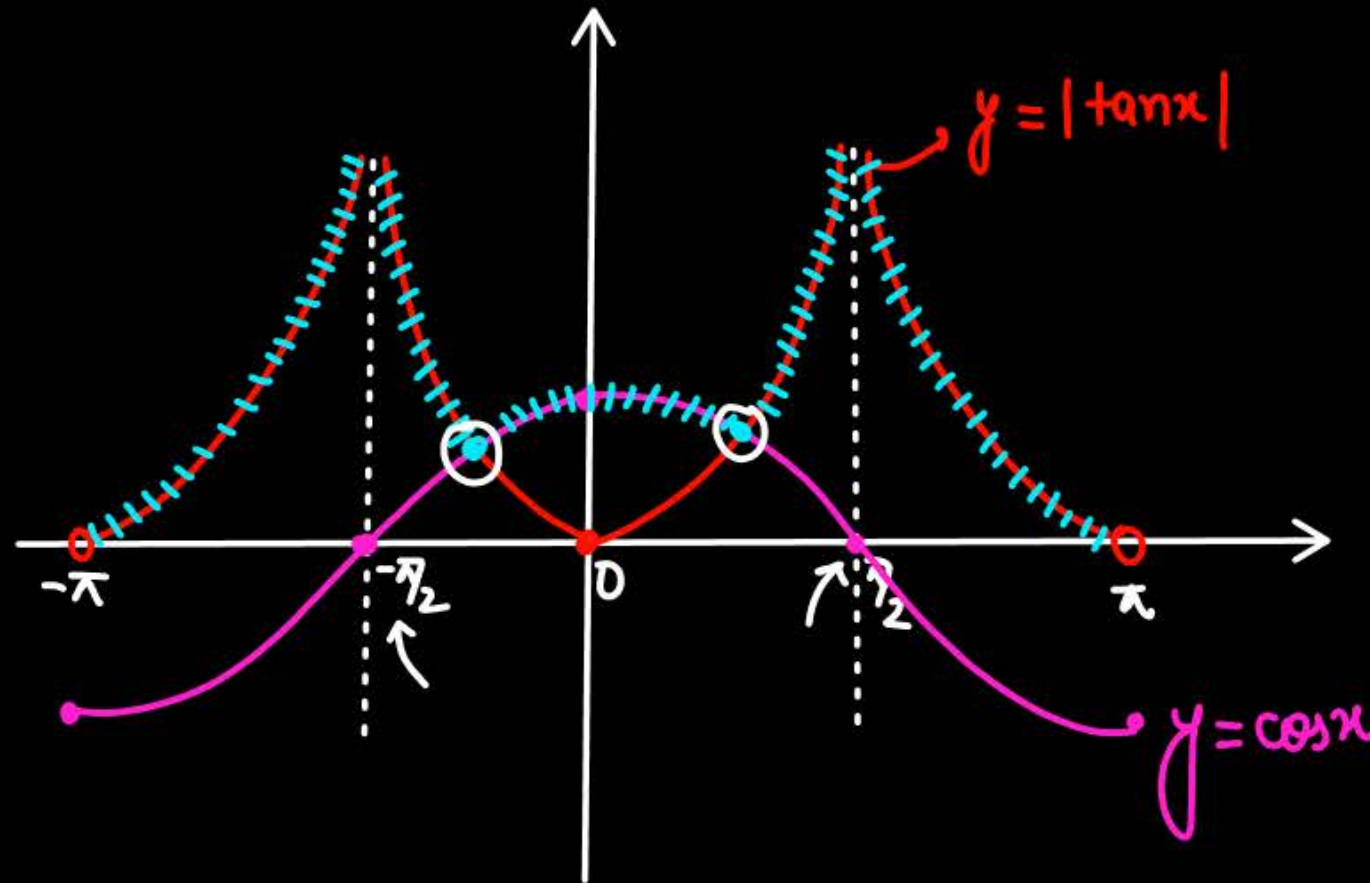
[IIT-JEE-2001]



Find number of points for which given functions are non-derivative:

(iii) $g(x) = \max.\{ |\tan x|, \cos|x| \}, x \in (-\pi, \pi)$

$\swarrow$
 $\cos x$



NOPDC = 2.

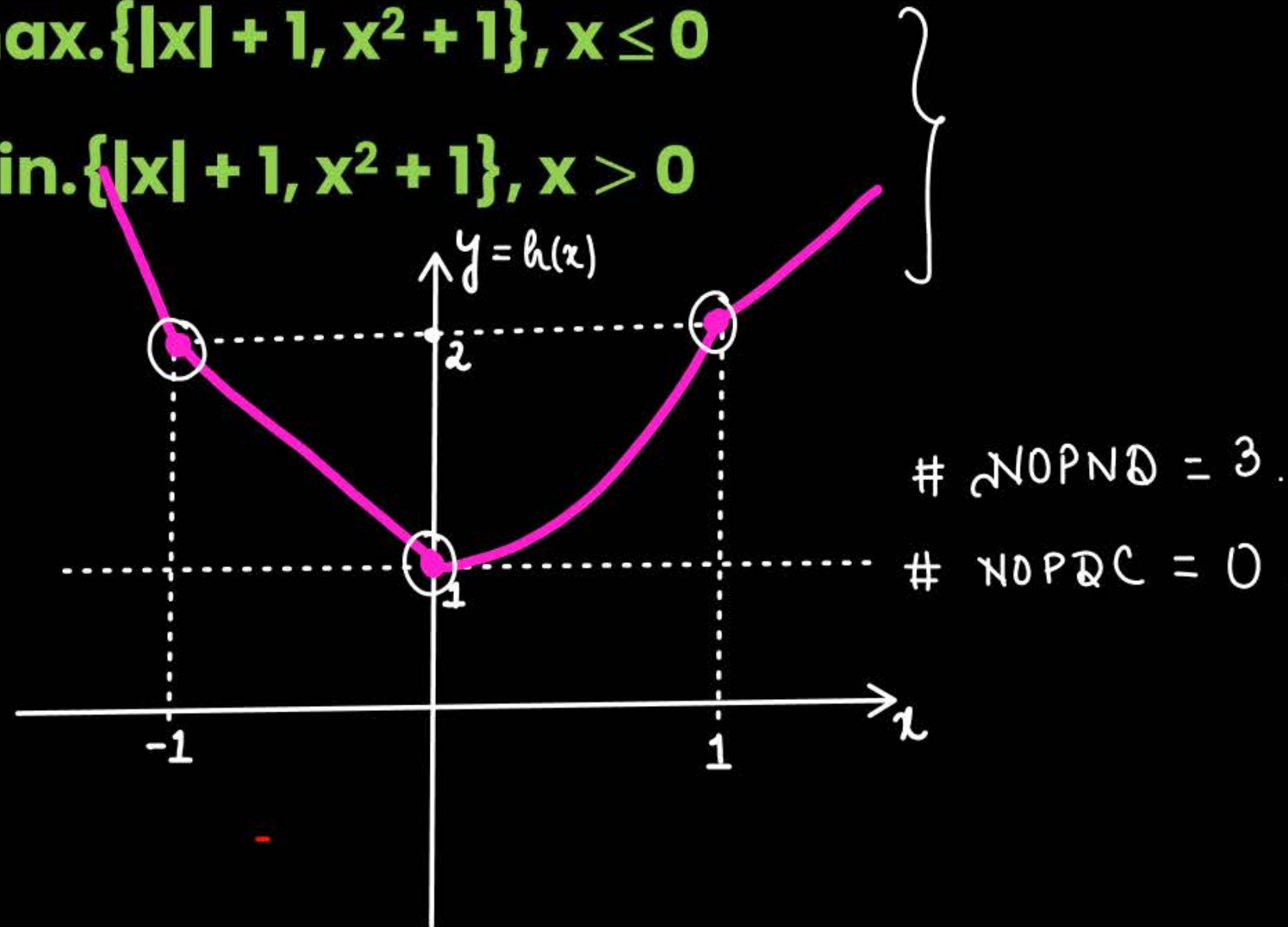
NOPNQ = 4.

Find number of points for which given functions are non-derivative:

(iv) $h(x) = \max.\{|x| + 1, x^2 + 1\}, x \leq 0$

$= \min.\{|x| + 1, x^2 + 1\}, x > 0$

[JEE (Adv.)-2014]





$$f(x) = \begin{cases} |1 - 4x^2|, & 0 \leq x < 1 \\ [x^2 - 2x], & 1 \leq x \leq 2 \end{cases}$$

Check the differentiability in $(0, 2)$ where $[]$ denotes greatest integer function.

→ Graph.

HW.

Let $f(x) = \begin{cases} \frac{\sin(x-[x])}{x-[x]}, & x \in (-2, -1) \\ \max\{2x, 3[|x|]\}, & |x| < 1 \\ 1, & \text{otherwise} \end{cases}$ & at $x = 1, -1 \Rightarrow f(x) = 1$, where

$[t]$ denotes greatest integer $\leq t$. Then the number of points where f is not continuous is _____.

$$\textcircled{\#} \text{ NO? } \text{NO} = 3.$$





ALGEBRA OF DIFFERENTIABILITY

Page No. 241



Exactly same as Continuity

D $\equiv$ differentiable function

N.D. $\equiv$ non-differentiable function

R-1: Sum or difference of two differentiable functions $\Rightarrow D_1 \pm D_2 =$ always diff.

R-2: Product of two differentiable function $\Rightarrow D_1 \times D_2 =$ always diff.

R-3: Division of two differentiable function $\Rightarrow \frac{D_1}{D_2} =$ always diff. (where $D_2 \neq 0$)



ALGEBRA OF CONTINUITY

Page No. 241



R-4 : Sum & difference of a differentiable & non-differentiable functions

$$\Rightarrow (D) \pm (N.D.) = \text{non-differentiable}$$

$$D \times (N.D.) \quad \frac{D}{N.D.} \quad \frac{N.D.}{D} \rightarrow \text{Check!}$$

R-5 : Sum, difference, product & division of two non-differentiable function
(Can't comment exactly before knowing function)

Check krna padega!!



Which of the following functions is differentiable at $x = 0$?

~~A~~ $\cos(|x|) + |x|$ $= \underbrace{\cos x}_{D.} + \underbrace{|x|}_{N.D.} \rightarrow \text{N.D. at } x=0$

~~B~~ $\cos(|x|) - |x|$

~~C~~ $\sin(|x|) + |x|$ $\xrightarrow{\text{N.D.}}$

~~D~~ $\sin(|x|) - |x|$ $\xrightarrow{\text{N.D.}}$

$$y = \sin x + x, \quad x \geq 0$$

$$= -(\sin x + x), \quad x < 0$$

at $x=0$

$$\text{RHD} = 2$$

$$\text{LHD} = -2$$

n.d. at $x=0$

$$y' = \cos x + 1, \quad x > 0$$

$$= -(\cos x + 1), \quad x < 0$$

$$y = \sin x - x, \quad x \geq 0$$

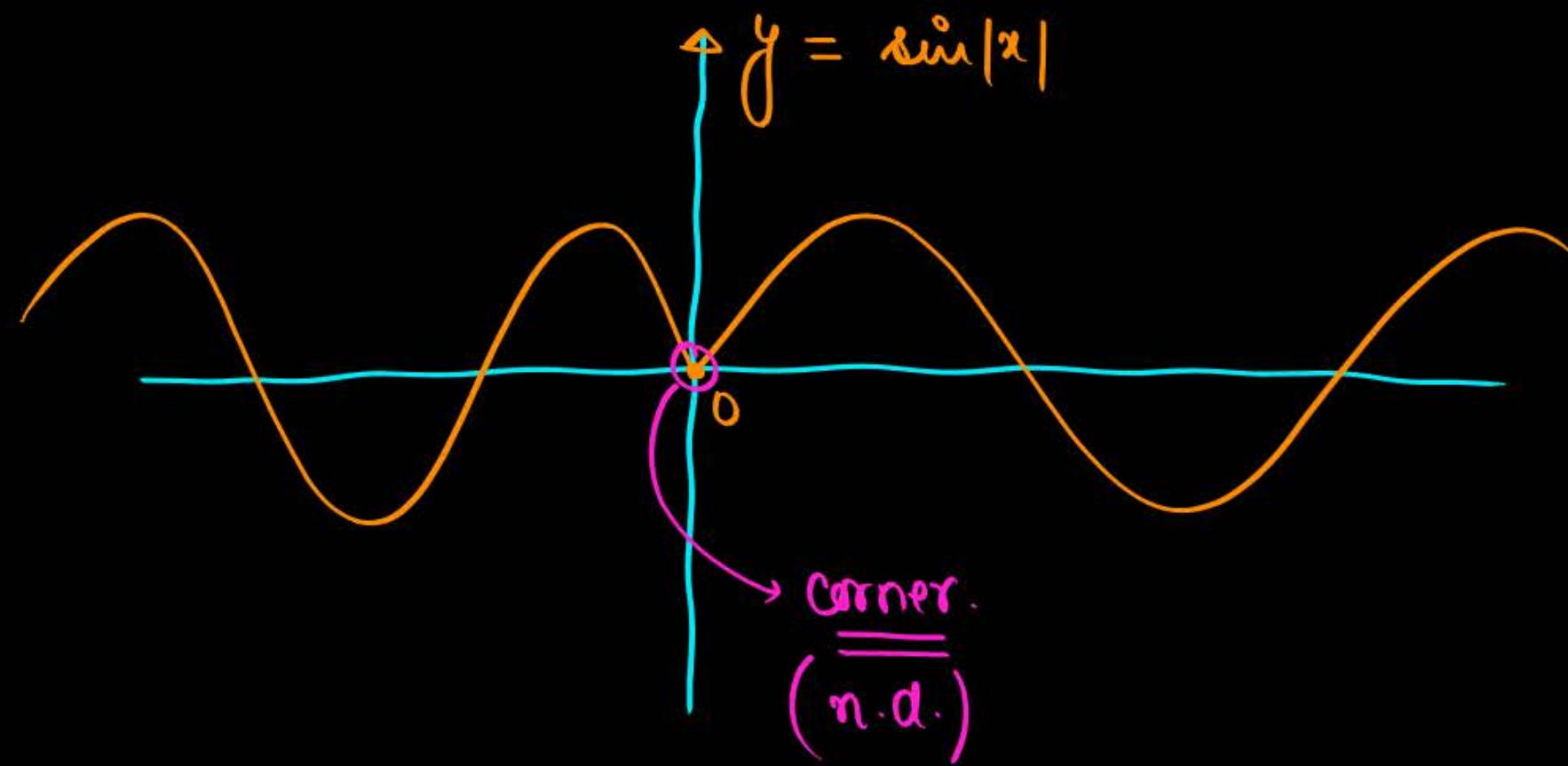
$$= -\sin x + x, \quad x < 0$$

$$y' = \cos x - 1, \quad x > 0$$

$$= -\cos x + 1, \quad x < 0$$

$$\text{RHD} = 0$$

$$\text{LHD} = 0$$





KHATARNAAK RESULTS

Page No. 242



Result-01: For multiplication of two function

At $x = a$;

If $f(x)$ is differentiable at $x = a$ with $f(a)=0$ & function $g(x)$ is continuous

Then: $y = f(x).g(x)$ is **always differentiable at $x = a$.**

diff. at $x=a$, $f(a)=0$

$$t(x) = \underbrace{f(x)}_{\text{diff. at } x=a, f(a)=0} \cdot \underbrace{g(x)}_{\text{cont. at } x=a}$$

cont. at $x=a$.

at $x=a$.

$$\text{RHD} = \lim_{h \rightarrow 0^+} \frac{f(a+h) \cdot g(a+h) - \cancel{f(a) \cdot g(a)}}{h} = \lim_{h \rightarrow 0^+} \left(\frac{f(a+h) - f(a)}{h} \right) g(a+h) = f'(a^+) \cdot g(a^+)$$

$$\text{LHD} = \lim_{h \rightarrow 0^+} \frac{f(a-h) \cdot g(a-h) - \cancel{f(a) \cdot g(a)}}{-h} = \lim_{h \rightarrow 0^+} \left(\frac{f(a-h) - f(a)}{-h} \right) g(a-h) = f'(a^-) \cdot g(a^-)$$

Check differentiability of given function over given interval :

(i) $f(x) = (\ln x) \cdot |x - 1|$ at $x = 1 \rightarrow \text{diff.} \checkmark$

(ii) $f(x) = (\sin \pi x) |(x - 1) \cdot (x - 2) \cdot (x - 3)|$ at $x = 1, 2, 3. \rightarrow \text{diff.} \checkmark$

(iii) $f(x) = |(x - 3)^7| + |(x - 7)^4| + |x - 5| \forall x \in \mathbb{R}.$

$\rightarrow$ (i) $f(x) = (\ln x) \cdot |x - 1|$, at $x = 1.$

$\downarrow$ $\downarrow$
 $\text{diff.} \checkmark$ cont.
 $\text{value} = 0$
 $\text{at } x = 1$

(ii) $f(x) = (\sin \pi x) \cdot \overbrace{|(x - 1)(x - 2)(x - 3)|}^{\text{cubic.}}$

$\downarrow$ $\downarrow$
 diff. cont.
 $\text{value} = 0$
 $\text{at } x = 1, 2, 3.$

Check differentiability of given function over given interval :

(i) $f(x) = (\ln x) \cdot |x - 1|$ at $x = 1$

(ii) $f(x) = (\sin \pi x) |(x - 1) \cdot (x - 2) \cdot (x - 3)|$ at $x = 1, 2, 3$.

(iii) $f(x) = |(x - 3)^7| + |(x - 7)^4| + |x - 5| \forall x \in \mathbb{R}$.

$$f(x) = \underbrace{|(x-3)^7|}_{\text{diff. always}} + \underbrace{|(x-7)^4|}_{\text{n.d. at } x=5} + \boxed{|x-5|}$$

n.d. only at $x=5$.

$$\underbrace{|(x-7)^4|}_{\text{+ve/0}} = (x-7)^4$$

$$y = |(x-3)^7|$$

at $x=3$

$$y = (x-3)^7, \quad x \geq 3$$

$$= -(x-3)^7, \quad x < 3$$

$$\left. \begin{aligned} \frac{dy}{dx} &= 7(x-3)^6, & x > 3 \\ &= -7(x-3)^6, & x < 3 \end{aligned} \right\}$$

at $x=3$

$$\hookrightarrow \text{LHD} = \text{RHD} = 0$$



KHATARNAAK RESULTS

Page No. 243



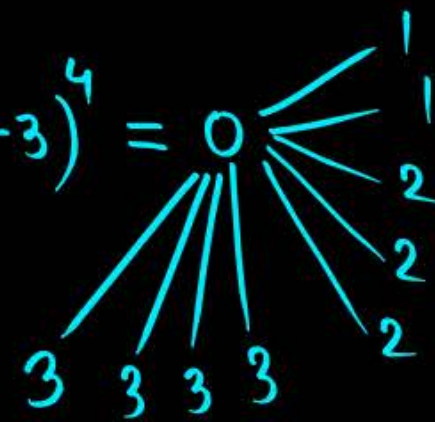
Result-02: For Repeated roots

If $x = a$ is repeated root of $f(x)$ then $f(x)$ is differentiable at $x = a$

Repeated Root:

$$f(x) = (x-1)(x-2)(x-3) = 0$$


$$f(x) = (x-1)^2(x-2)(x-3) = 0$$


$$f(x) = (x-1)^2(x-2)^3(x-3)^4 = 0$$


Let $S = \{t \in \mathbb{R} : f(x) = |x - \pi| \cdot (e^{|x|} - 1) \sin|x| \text{ is not differentiable at } t\}$.
Then the set S is equal to.

$$S = \{ \}$$

$$f(x) = \underbrace{|x - \pi|}_{\substack{x = \pi \\ \text{root}}} \cdot \underbrace{(e^{|x|} - 1)}_{\substack{x = 0 \\ \text{root}}} \cdot \underbrace{\sin|x|}_{\substack{x = 0, \pi \\ \text{roots}}}$$

$x = 0, x = \pi$ is repeated root.

S.P. $\Rightarrow x = 0, \pi$
 $\downarrow$
diff. ✓

☒ A ϕ (an empty set)

☐ B $\{0\}$

☐ C $\{\pi\}$

☐ D $\{0, \pi\}$

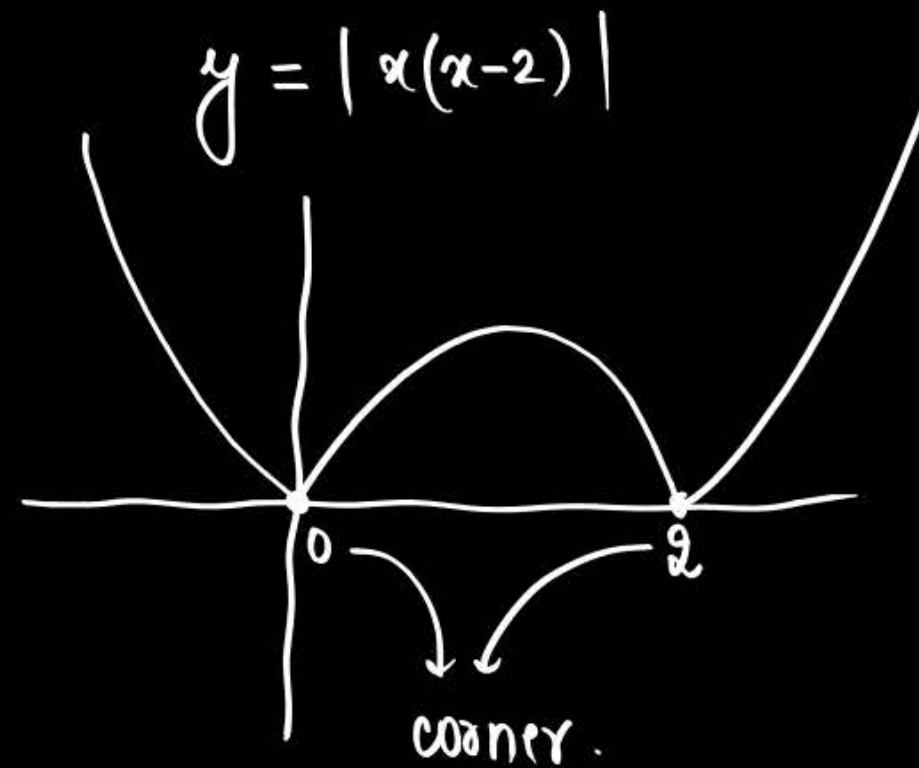
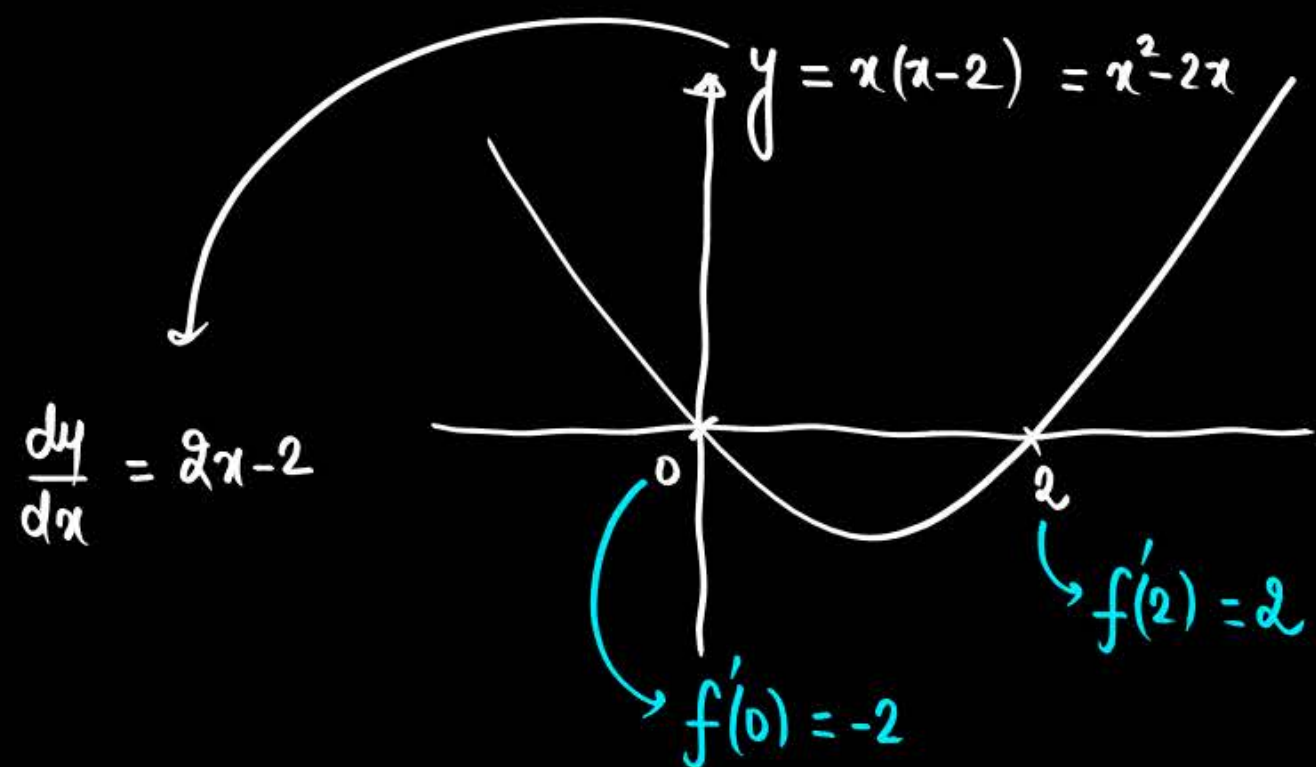
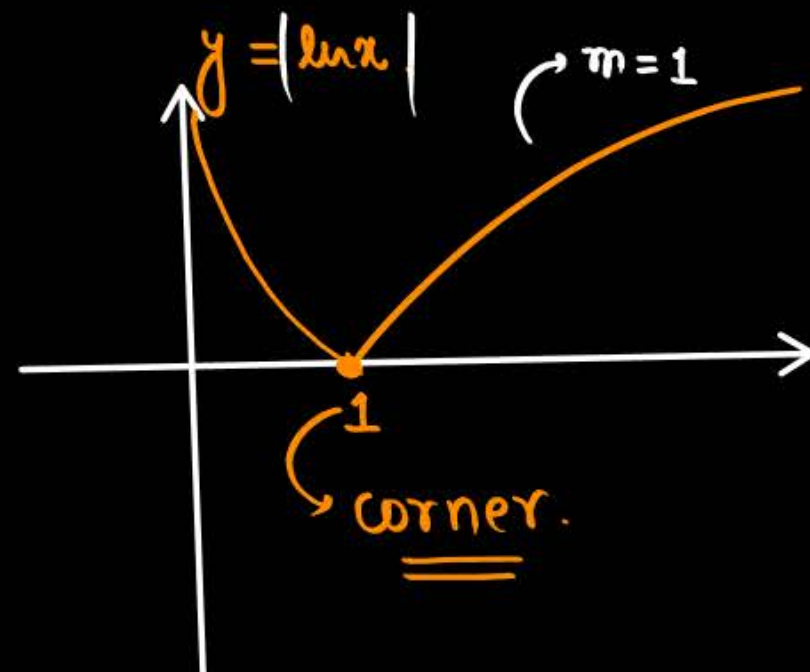
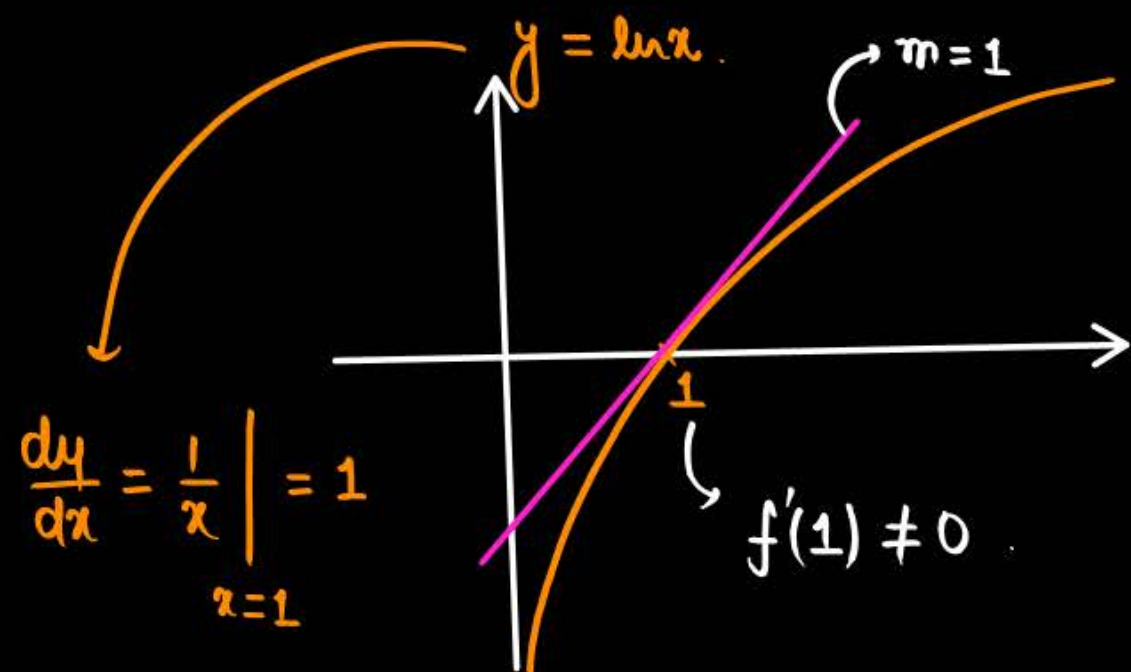


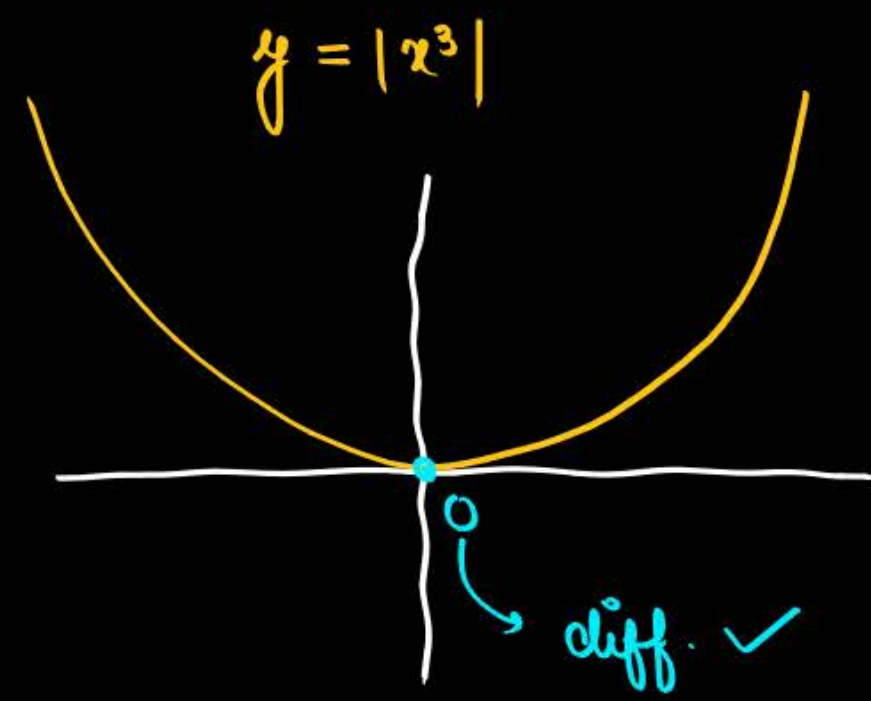
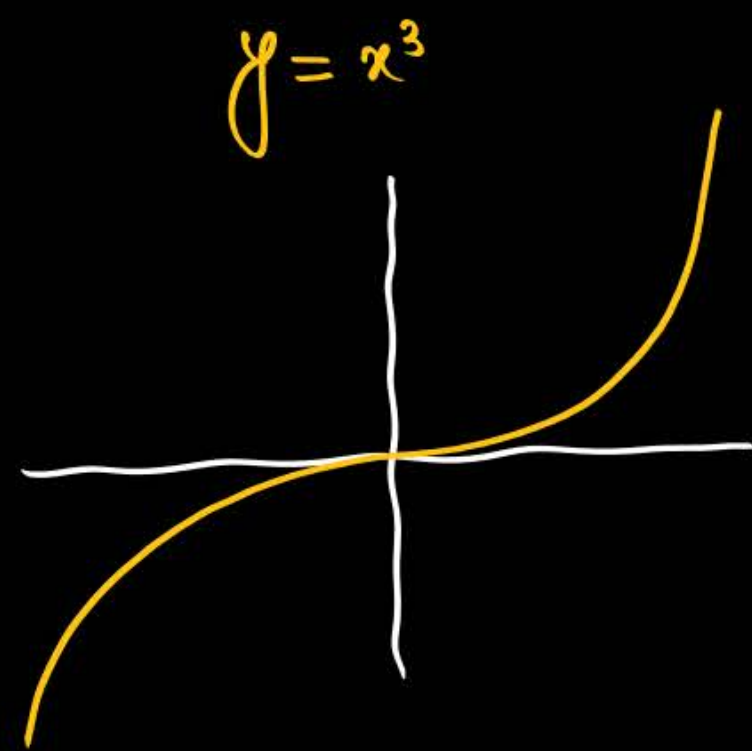
KHATARNAAK RESULTS



Result-03:

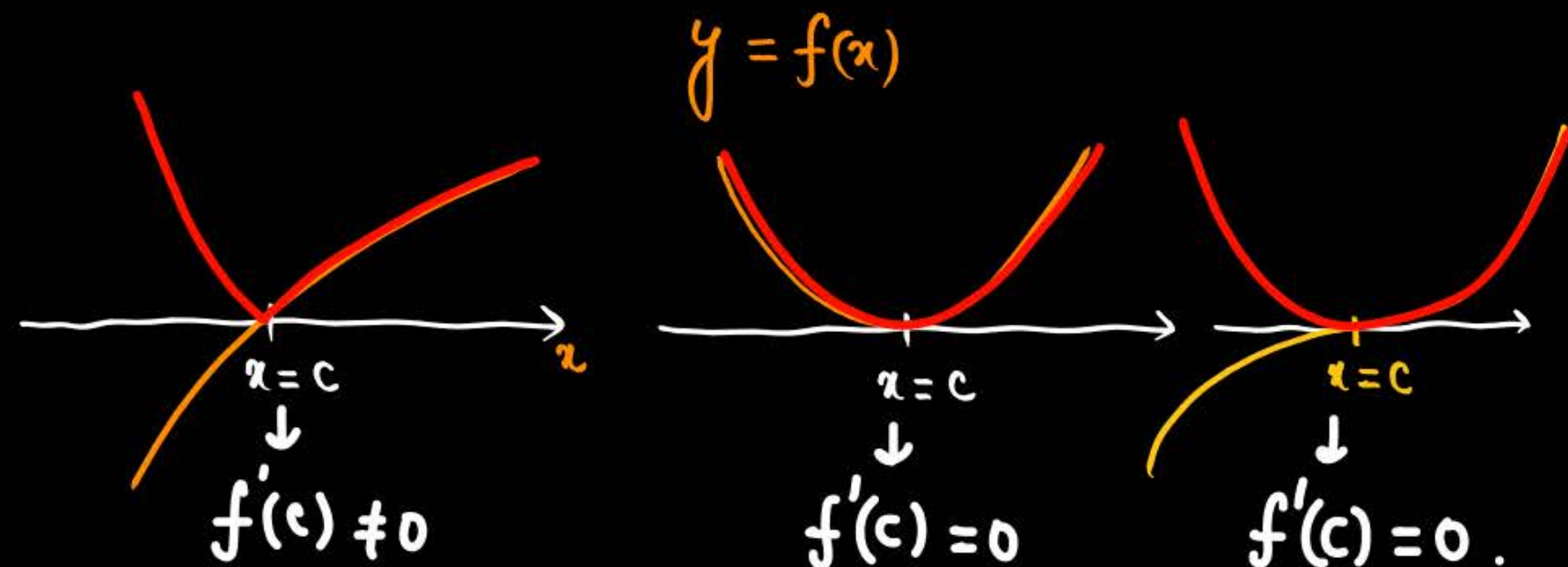
If $f(x)$ crosses X -axis at $x = a$ and $f'(a) \neq 0$,
then $|f(x)|$ is non-differentiable at $x = a$.





Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable at $c \in \mathbb{R}$ and $f(c) = 0$. If $g(x) = |f(x)|$, then at $x = c$, g is

- ☒ A Not differentiable if $f'(c) = 0$
- ☐ B Differentiable if $f'(c) = 0$
- ☒ C Not differentiable
- ☐ D Not differentiable if $f'(c) \neq 0$



Question [JEE (Adv.)–2020, Paper–1]



Let the function $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^3 - x^2 + (x - 1)\sin x$ and let $g: \mathbb{R} \rightarrow \mathbb{R}$ be an arbitrary function. Let $fg: \mathbb{R} \rightarrow \mathbb{R}$ be the product function defined by $(fg)(x) = f(x)g(x)$. Then which of the following statements is/are TRUE?

Hint

- A** If g is continuous at $x = 1$, then fg is differentiable at $x = 1$
- B** If fg is differentiable at $x = 1$, then g is continuous at $x = 1$
- C** If g is differentiable at $x = 1$, then fg is differentiable at $x = 1$
- D** If fg is differentiable at $x = 1$, then g is differentiable at $x = 1$



KHATARNAAK RESULTS

Page No. 243



Result-04: (Major Reason of Non-diff.) ★★★★★★★★★★
★★★★★★★★★★

If $h(x) = f(x) \cdot g(x)$, where $f(x)$ is differentiable at $x = a$ with $f(a) \neq 0$ and $g(x)$ is continuous but non-differentiable at $x = a$ & $g(a) = 0$.

Then $h(x)$ is non-differentiable at $x = a$.

$$h(x) = \underbrace{f(x)}_{\substack{\text{diff. } \checkmark \\ f(a) \neq 0}} \cdot \underbrace{g(x)}_{\substack{\text{c \& n.d.} \\ g(a) = 0}}$$

R-1.

$$h(x) = \underbrace{f(x)}_{\substack{\text{diff. } \checkmark \\ f(a)=0}} \cdot \underbrace{g(x)}_{\text{cont.}} \rightarrow \text{diff. at } x=a$$

R-4.

$$h(x) = \underbrace{f(x)}_{\substack{\text{diff. } \checkmark \\ f(a) \neq 0}} \cdot \underbrace{g(x)}_{\substack{\text{c \& n.d.} \\ g(a)=0}} \rightarrow \text{non-diff. at } x=a$$

The function $f(x) = |x^2 - 2x - 3| \cdot e^{|9x^2 - 12x + 4|}$ is not differentiable at exactly

A four points

B three points

☒ C two points

D one point

$$\# f(x) = |(x-3)(x+1)| \cdot e^{(3x-2)^2}$$

$\downarrow$
 C & non-diff. at $x=3$.
 Value = 0.

$\downarrow$
 diff. at $x=3$
 value $\neq 0$.

S.P. $\rightarrow$ $x=3, -1$

$\swarrow$ n.d. $\searrow$ n.d.

IMPORTANT NOTE: DUMMY MOD SIGN



1. $|(x - \alpha)^{\text{even}}| = (x - \alpha)^{\text{even}}$
2. $\cos|x| = \cos x$
3. $|x \sin x| = x \sin x$, $x \in (-\pi, \pi)$
4. $|x| \cdot \sin|x| = x \sin x$

} "Hint"

Question [JEE Mains-2022]



The number of points, where the function

$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = |x-1|\cos|x-2|\sin|x-1| + |x-3|\underbrace{|x^2-5x+4|}_{|(x-1)(x-4)|}$, is **NOT** differentiable, is

A 1

B 2

C 3

D 4

$$f(x) = \underbrace{\cos(x-2)}_{\# \text{ diff. always}} \underbrace{(x-1)}_{\# \text{ diff. always}} \underbrace{\sin(x-1)}_{\# \text{ diff. always}} + |x-3| \cdot |x-1| \cdot |x-4|$$

diff. always

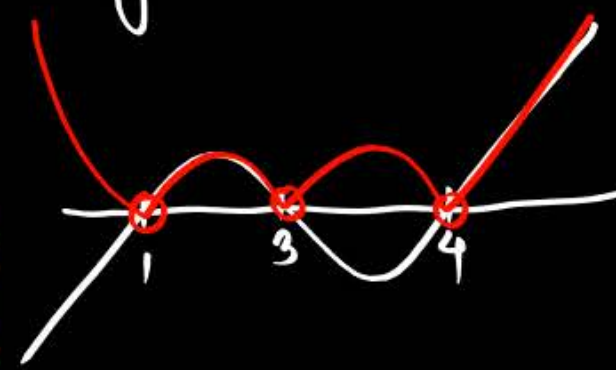
$$|(x-1)(x-3)(x-4)|$$

Result

n.d. at $x=1$ $\left(\begin{array}{l} \underbrace{|(x-3)(x-4)|}_{\text{diff at } x=1} \cdot \underbrace{|x-1|}_{\text{c + n.d. value } = 0} \\ \text{value } \neq 0 \end{array} \right)$

cubic

$$y = (x-1)(x-3)(x-4)$$



Question [JEE-1999, 2M]



The function $f(x) = (x^2 - 1)|x^2 - 3x + 2| + \cos(|x|)$ is not differentiable at

A -1

B 0

C 1

D 2



Question [JEE (Adv.)–2016, Paper–1]

Let $a, b \in \mathbb{R}$ and $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by
 $f(x) = a\cos(|x^3 - x|) + b|x|\sin(|x^3 + x|)$. Then f is

- A** Differentiable at $x = 0$ if $a = 0$ and $b = 1$
- B** Differentiable at $x = 1$ if $a = 1$ and $b = 0$
- C** NOT differentiable at $x = 0$ if $a = 1$ and $b = 0$
- D** NOT differentiable at $x = 1$ if $a = 1$ and $b = 1$

Ans



Check differentiability of following :

(i) $f(x) = \begin{cases} 3^x, & x \in [-1, 1] \\ 4 - x, & x \in [1, 4] \end{cases}$ **at** $x = 1$.

DiBY!!

(ii) $f(x) = x \left(\frac{e^{1/x} - e^{-1/x}}{e^{1/x} + e^{-1/x}} \right), x \neq 0$ & $f(0) = 0$, **at** $x = 0$.

DIBY-A

Discuss differentiability of function

$$f(x) = \begin{cases} 2x - 2, & x < 1 \\ x^2, & 1 \leq x \leq 2 \\ -2 + 8x - x^2, & x > 2 \end{cases} \text{ at } x = 1 \text{ and } x = 2$$

DIBY-B

If $f(x) = \begin{cases} x^3 + 2x^2 + ax + 6, & x < 1 \\ 2x + b, & x \geq 1 \end{cases}$ is differentiable for all $x \in R$, then find the value of $(a + b)$

DIBY-C

Let $f(x) = |x - 2|e^x g(x) + \sin x$, where $g(x)$ is a continuous function. If $f(x)$ is differentiable at $x = 2$, then find $g(2)$

DIBY-D

Let $f(x) = \begin{cases} x^2 \left| \cos \frac{\pi}{2x} \right|, & x \neq 0 \\ 0, & x = 0 \end{cases}$. Find $f' \left(\frac{1}{3}^+ \right) + f' \left(\frac{1}{3}^- \right)$.

DIBY-E

$f(x) = \lim_{x \rightarrow \infty} \frac{x^n - \sin x^n}{x^n + \sin x^n} x > 0; x \neq 1, f(1) = 0$ discuss differentiability at $x = 1$

DIBY-F

$f(x) = \begin{cases} \left\{ x + \frac{1}{3} \right\} [\sin \pi x] & 0 \leq x < 1 \\ [2x] \operatorname{sgn}(x - 4/3) & 1 \leq x \leq 2 \end{cases}$ check differentiability at $x = 1$ where $[\cdot]$ is GIF
 $\{\cdot\}$ is fpf

DIBY-G

$f(x) = \begin{cases} x^2 \operatorname{sgn}[x] + \{x\} & 0 \leq x < 2 \\ \sin x + |x - 3| & 2 \leq x \leq 4 \end{cases}$. Check differentiability at $x = 2$ where $[]$ is GIF:
 $\{ \}$ is fpf

DIBY-H

Let $f(x) = \begin{cases} \frac{5e^{\frac{1}{x}} + 2}{3 - e^{\frac{1}{x}}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

Check (i) $f(x)$ (ii) $x^2 f(x)$ for continuity and differentiability at $x = 0$.

DIBY-I

Let a_1 and a_2 be two values of a for which

$f(x) = \begin{cases} x \cdot \left(\frac{\ln(1+x) + \ln(1-x)}{\sec x - \cos x} \right), & x \in (-1, 0) \\ (a^2 - 3a - 1)\sin x + x^2, & x \in [0, \infty) \end{cases}$ is differentiable at $x = 0$, then find the value of $(a_1^2 + a_2^2)$.

DIBY-J

Let $f(x) = \begin{cases} x \tan^{-1} x + \sec^{-1} \frac{1}{x}, & x \in (-1, 1) - \{0\} \\ \frac{\pi}{2}, & x = 0 \end{cases}$ If $I = f'(0^+)$ and $m = f'(0^-)$, then find the value of $I^2 + m^2$.

DIBY-K

Let $g(x) = \begin{cases} \frac{x^2 + x \tan x - x \tan 2x}{ax + \tan x - \tan 3x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$. If $g'(0)$ exists and is equal to non zero value b , then find $\frac{b}{a}$.

DIBY-L (Graph)

$$f(x) = \begin{cases} \max \{ \sqrt{1+x^2}, \sqrt{4-x^2} \}, & -2 \leq x \leq 0 \\ \min \{ \sqrt{1+x^2}, \sqrt{4-x^2} \}, & 0 < x \leq 2 \end{cases}$$

Discuss continuity and differentiability of $f(x)$.

DIBY-M

Find the number of values of x in $[-4, 4]$ where

$f(x) = [3x + 14] + |4x^2 - 1|(2x^2 + 3x - 2) + \sin\left(\frac{\pi x}{2}\right)$ is non-derivable where $[.]$ is GIF.

DIBY-N

$f(x) = \begin{cases} |x^5 - 4x^3 + 2|\cos \pi x & x \geq 0 \\ x^3 - \sin(x) - |(x^2 + 8x + 12)x^2| & x < 0 \end{cases}$ number of points where $f(-|x|)$ is

not differentiable.

DIBY-O (Graph)

If n_1 = number of points of non-derivability of

$f(x) = \begin{cases} \max. \{|x|, \sin |x|\}, & x \in \left(0, \frac{\pi}{2}\right) \\ \max. \{|x|, \tan |x|\}, & x \in \left(\frac{-\pi}{2}, 0\right) \end{cases}$ in $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$ and n_2 = number of points of

discontinuity of $g(x) = [\sin x] + [\cos x]$ where $[.]$ is GIF in $[0, 2\pi]$ then find $(n_1 + n_2)$

RAPID FIRE DIBY's (ANSWERS)



- (A) **Non-differentiable at $x = 1, 2$**
- (B) **-3**
- (C) **0**
- (D) **0**
- (E) **Non-differentiable**
- (F) **Non-differentiable**
- (G) **Non-differentiable**
- (H) (i) **Discontinuous**
(ii) **Continuous & differentiable**

- (I) **9**
- (J) **2**
- (K) **$\frac{7}{52}$**
- (L) **Non-differentiable at $x = 0, 2, \pm\sqrt{\frac{3}{2}}$**
- (M) **Non-derivable at 25 points**
- (N) **5**
- (O) **6**

DO IT BY YOURSELF (DIBY)

CONTINUITY



DIBY-16

The number of values of x , $x \in [-2, 3]$ where $f(x) = [x^2]\sin\{\pi x\}$ is discontinuous is (where $[.]$ denotes greatest integer function)

[Ans. : 8]

DIBY-17

Let $[x]$ be the greatest integer less than or equals to x . Then, at which of the following point(s) the function $f(x) = x \cos(\pi(x + [x]))$ is discontinuous?

[IIT-JEE 2012]

- (A) $x = -1$ (B) $x = 1$ (C) $x = 0$ (D) $x = 2$

[Ans. : (A, B, D)]

DIBY-18

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that $f(3x) - f(x) = x$. If $f(8) = 7$, then $f(14)$ is equal to:

[JEE Mains-2022]

- (A) 4 (B) 10 (C) 11 (D) 16

[Ans. : (B)]

DIBY-19



If the function $f(x) = \begin{cases} (1 + |\cos x|)^{\frac{\lambda}{|\cos x|}}, & 0 < x < \frac{\pi}{2} \\ \mu, & x = \frac{\pi}{2} \\ e^{\frac{\cot 6x}{\cot 4x}}, & \frac{\pi}{2} < x < \pi \end{cases}$ is continuous at $x = \frac{\pi}{2}$,

then $9\lambda + 6\log_e \mu + \mu^6 - e^{6\lambda}$ is equal to

[JEE Mains-2023]

- (A) 11 (B) 8 (C) $2e^4 + 8$ (D) 10

Ans.: (D)

DIBY-20

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function as $f(x) = \begin{cases} 5, & \text{if } x \leq 1 \\ a + bx, & \text{if } 1 < x < 3 \\ b + 5x, & \text{if } 3 \leq x < 5 \\ 30, & \text{if } x \geq 5 \end{cases}$ Then, f is [JEE Mains-2019]

- (A) Continuous if $a = -5$ and $b = 10$ (B) Continuous if $a = 5$ and $b = 5$
(C) Continuous if $a = 0$ and $b = 5$ (D) Not continuous for any values of a and b

Ans.: (D)



Homework



Re-attempt all good Questions of Class

DIBY

*It is never about end result
It's always about Journey*



THANK
You

#futureellTians